Master of Data Science and Innovation

36118 Applied Natural Language Processing (ANLP)

Session 8

Dr. Antonette Shibani

Senior Lecturer, MDSI teaching team

Let’s start with a check-in

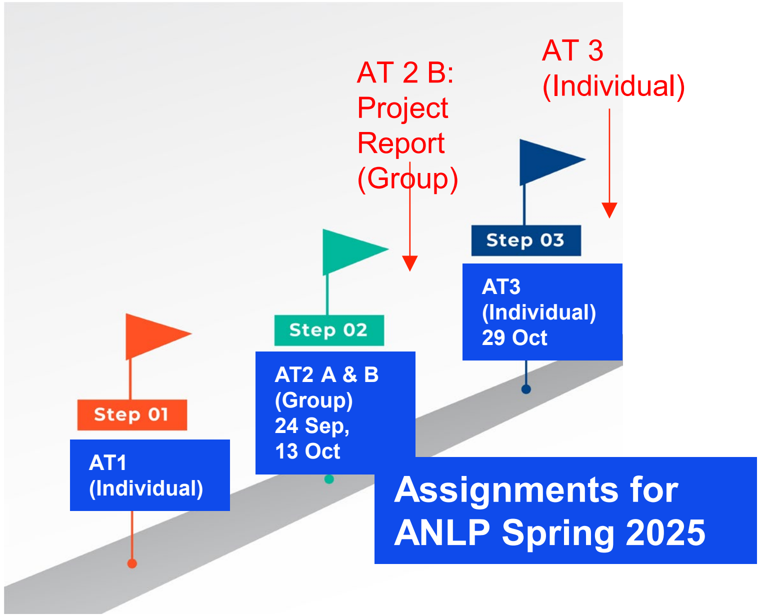
Menti Link provided in class

Update on Assignments

Well done on AT2 a, b – You've crossed the second milestone (almost!).

Complete the self and peer review component for AT2B on SPARK (Don't be too strict on yourself too, as it may affect your weightage factor)

Only one more to go ☺



Nominate your team for Project Awards (due this Weds, 15 Oct)

AT2 Awards will also be assessed by an external judge (and we want to give them some info prior to the session)

Any one student from the team can nominate by filling the form below:

ANLP Project Nomination for Best Project Awards (Responses Due Wednesday, 15 Oct 2025)

Any one member from a team can nominate their project work to compete for the Best Project Award for ANLP Spring 2025.
Eligibility criteria is below:

- All team members should have submitted AT2 Project Report on Canvas and and Peer Evaluation on SPARK (individually) by Wednesday, 15 Oct 2025
- The team should have created a working Python web application (a local app is fine, doesn't need to be hosted on the cloud). The work cannot just be in notebook format or scripts
- At least one team member should be ready to present the project pitch (3-minute summary) in the Final Showcase in class on Monday, 20 Oct 2025. We also expect all team members to be present at the showcase and networking event in this last session.

1. Your Name *

Enter your answer

2. Your Group number *

Enter your answer

3. Project Title *

Enter your answer

4. What do you think are some unique features of your app and why is your project work significant? Why should you win the award? (Answer in 1-3 sentences) *

Enter your answer

5. Have you met all eligibility conditions and agree to above requirements? *

○ Yes, I confirm

https://forms.office.com/r/DpBBhhFGVt

Project Showcase (Demo) and Networking Session: **Monday Oct 20**

- Last in-class session of the semester (Session 9) – we'll make it a celebration of your hard work!
- Attendance is mandatory for all students. Make sure all teams arrange a working demo of your projects and prepare the short presentation (team intro + 3-minute pitch)
- Posters can be displayed digitally on the screens, no need to print! We'll be printing some copies of posters from teams nominated for awards for the judges.
- A guest speaker/ judge will be attending this session

AT2 Showcase

- Each team will set up a demo in your laptops (mandatory for all students). Connect the working prototype to the table’s projector screen. Use the digital poster for a summary of your work.
- Teams will get exactly **3 minutes** to pitch their work (aim to present highlights only, do not have more than 5 slides/ speakers!). You may first introduce members of the team separately, helps with attendance-taking (not counted towards the 3-min limit)
- All PPTs should be uploaded in a shared folder at the start of the session

Ethics and fairness in NLP

What pops in your mind when you hear ‘Ethics’?

Menti poll

Ethics

https://www.merriam-webster.com/dictionary/ethics

Ethics

Ethics is a study of what are **good and bad** ends to pursue in life and what it is **right and wrong** to do in the conduct of life.

It is therefore, above all, a practical discipline.

Its primary aim is to determine how one ought to live and what actions one ought to do in the conduct of one's life."

Introduction to Ethics, John Deigh

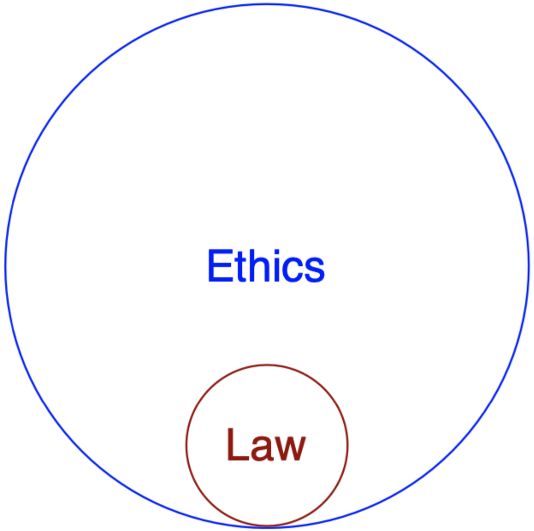
Ethics

It’s the **good** things

It’s the **right** things

But, how simple is it to define what’s good and what’s right?

Ethics is not law



Law: sets minimum standards (rules and regulations)

vs

Ethics: sets maximum standards

Sometimes what is legal may not be ethical!

Why should you care?

- because you are a citizen, not only a data professional/ researcher/ educator
- because if you don’t as specialists, others with less knowledge will

Ethics by design: *Not* an after-thought after development or something to delegate

Not through “techno-solutionism”

NLP: Language use is fundamentally a social activity

"The common misconception is that language has to do with words and what they mean. It doesn't. It has to do with **people** and what they mean."

We are trying to understand the speaker's intentions!

Herbert H. Clark & Michael F. Schober (1992)
Asking Questions and Influencing Answers

http://web.stanford.edu/~clark/1990s/Clark,%20H.H.%20_%20Schober,%20M.F.%20_Asking%20questions%20and%20influencing%20answers_%201992.pdf

Decisions we make about our data, methods, and tools are tied up with their *impact on people and societies*.

Ethics and fairness in NLP

- Core thread running through the curriculum for Human-centered data science and ethical thinking
- We’ve covered parts of these in almost every lecture!
- *Not* an after-thought after development or something to delegate

Some examples from prior sessions

- Understanding nuances in human language
- ML – the IQ dilemma
- Gender bias in vector embeddings
- Deep learning applications – implications on the real world
- …..

Nuanced ethical decisions often come with a

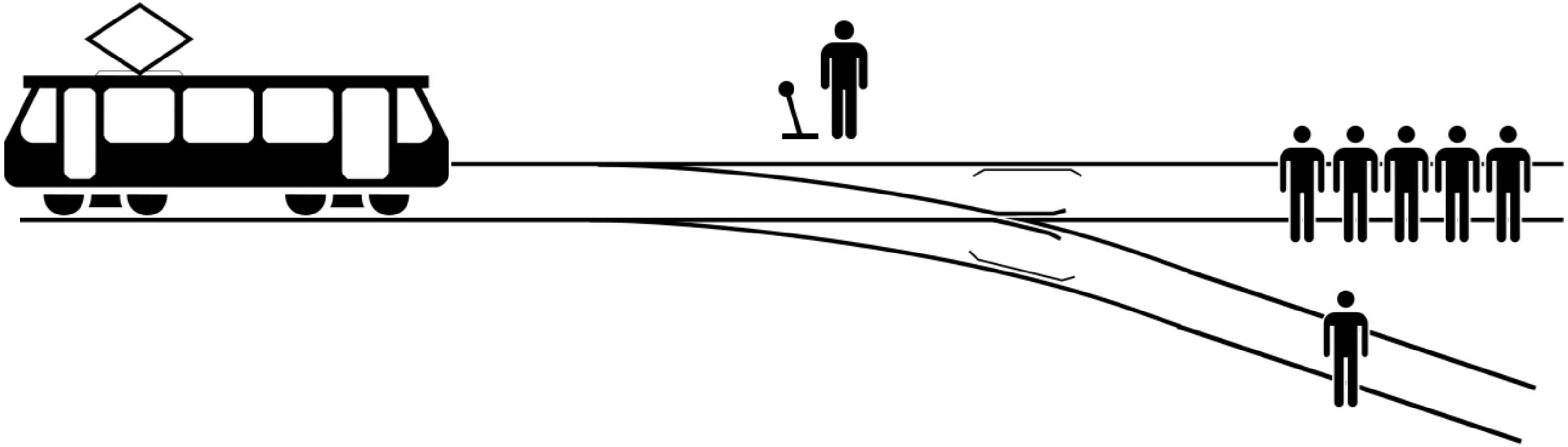
moral dilemma

Not always easy to conclude what's right or wrong!

Knight, S., Shibani, A., & Buckingham Shum, S. (2023). **A reflective design case of practical micro-ethics in learning analytics.** *British Journal of Educational Technology*, *54*(6), 1837-1857.

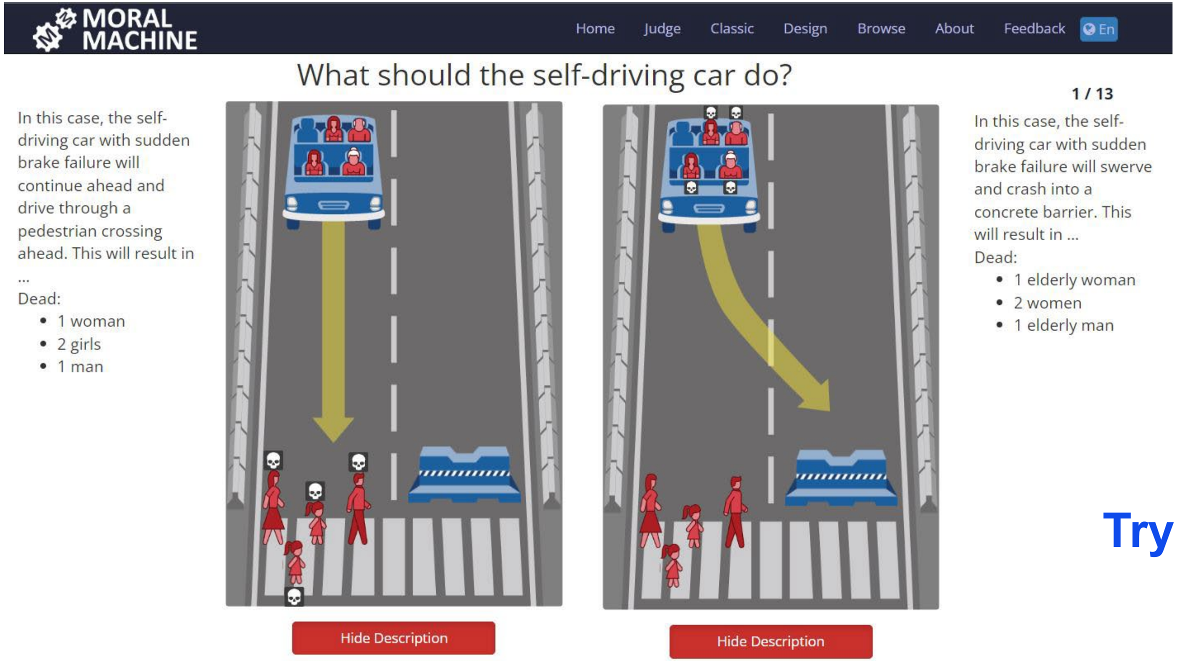
The Trolley Dilemma

Should you pull the lever to divert the trolley?



[image from Wikipedia]

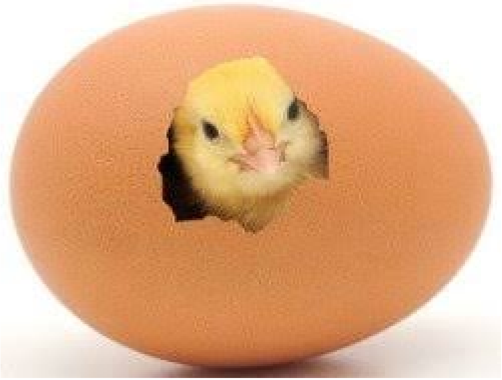
Moral machine

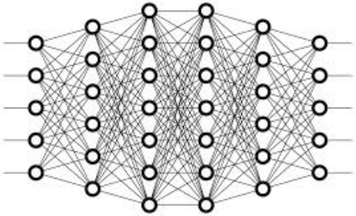


https://www.moralmachine.net/

Try this out!

The Chicken dilemma





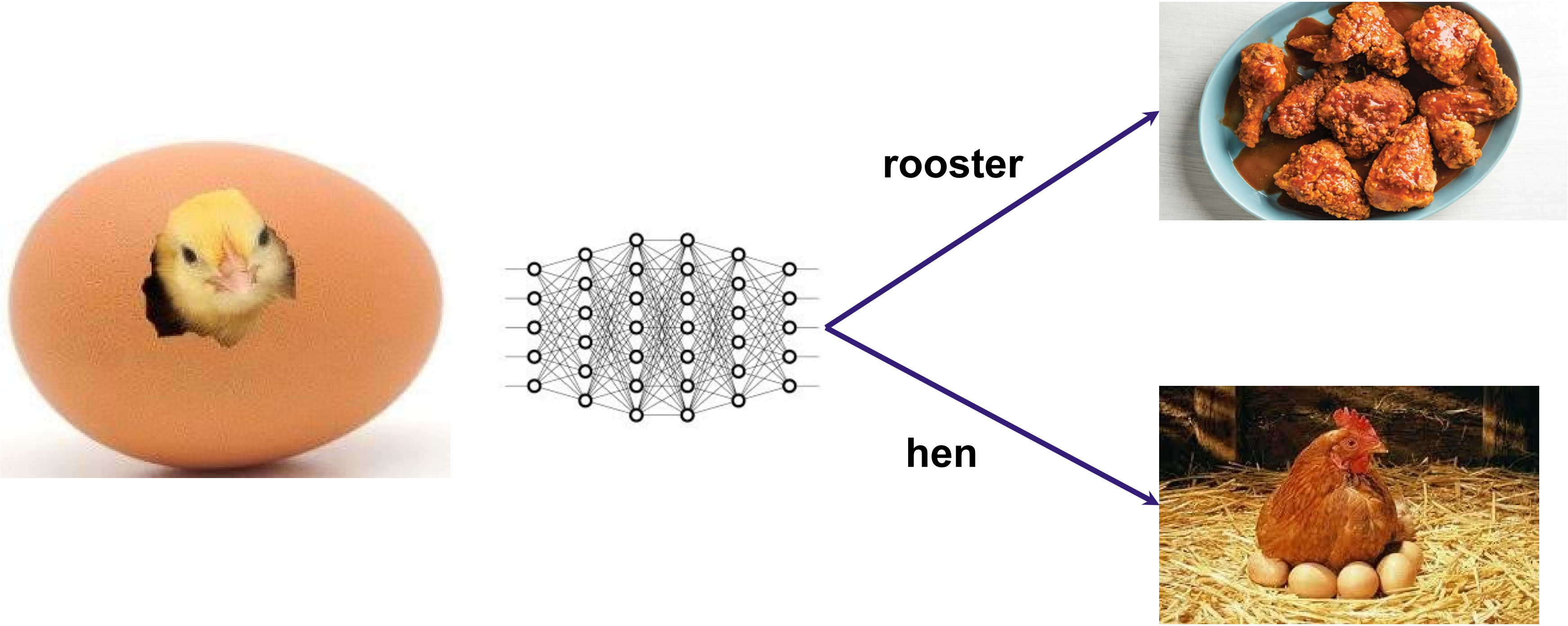
This chicken (male)
goes to a meat farm

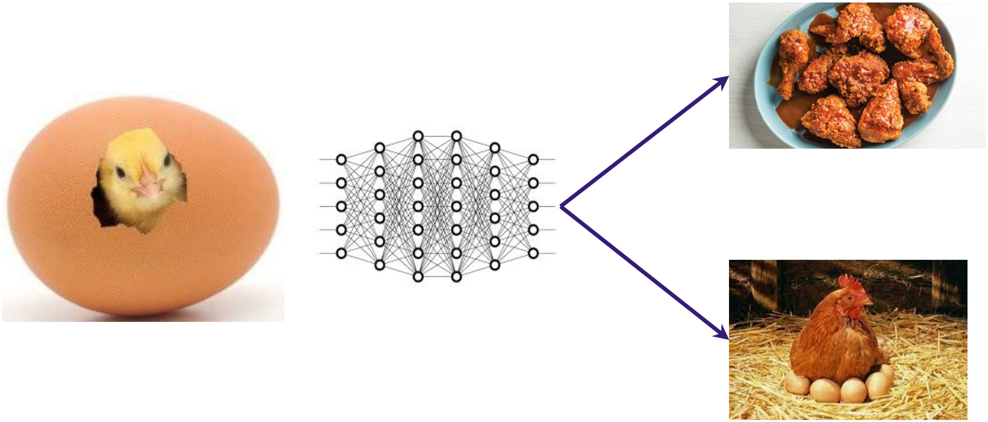
rooster

hen

This chicken (female)
goes to an egg farm

Is this Ethical?



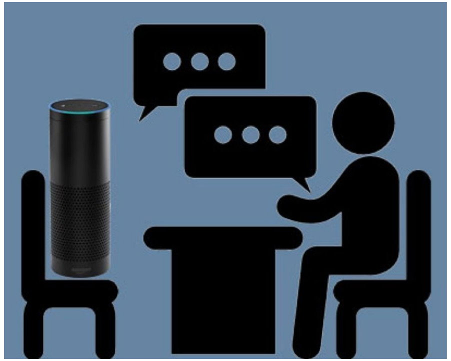


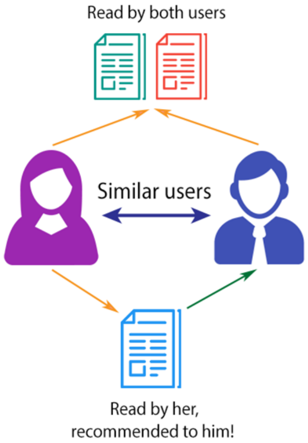
- ➔ Ethics is inner guiding, moral principles, and values of people and society
- ➔ There are gray areas. We often don’t have easy answers.
- ➔ Ethics changes over time with values and beliefs of people
- ➔ Legal ≠ Ethical

We will not give answers to such questions.
But we will learn how to reason about a technology and its (potential) impacts on people.

Group activity: AI and people

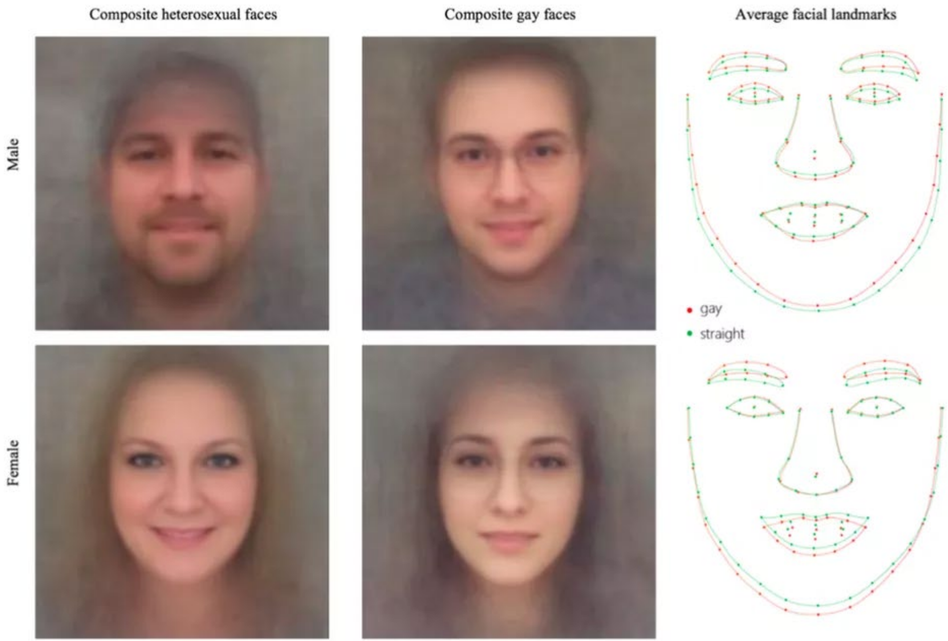
These AI systems make decisions about people!





Pick one example, discuss if it is cool or creepy (What are the arguments 'for' and 'against'?) Tell us why!

A recent study: the “AI Gaydar”, 2017



A recent study: the “AI Gaydar”

- Research question
  - Identification of sexual orientation from facial features
- Data collection
  - Photos downloaded from a popular American dating website
  - 35,326 pictures of 14,776 people, all white, with gay and straight, male and female, all represented evenly
- Method
  - A deep learning model was used to extract facial features + grooming features; then a logistic regression classifier was applied for classification
- Accuracy
  - 81% for men, 74% for women
- Motivation for the study: expose a threat to the privacy and safety of gay men and women

Extensively covered by press: https://www.nytimes.com/2017/10/09/science/stanford-sexual-orientation-study.html

Let’s discuss...

- Research question
  - Identification of sexual orientation from facial features
- Data collection
  - Photos downloaded from a popular American dating website
  - 35,326 pictures of 14,776 people, all white, with gay and straight, male and female, all represented evenly
- Method
  - A deep learning model was used to extract facial features + grooming features; then a logistic regression classifier was applied for classification
- Accuracy
  - 81% for men, 74% for women

What went wrong?

Questioning the ethics of the research question

- Identification of sexual orientation from facial features

How can people be harmed by this research?

Sexual orientation classifier - who can be harmed?

- In many countries being gay person is prosecutable (by law or by society) and in some places there is even death penalty for it
- It might affect people’s employment; family relationships; health care opportunities
- Personal attributes like gender, race, sexual orientation, religion are social constructs. They can change over time. They can be non-binary. They are private, intimate, often not visible publicly.
- Importantly, these are properties for which people are often discriminated against.

Dual framing in predictive analytics

Predictive policing is a real thing!

“We live in a dangerous world, where harm doers and criminals easily mingle with the general population; the vast majority of them are unknown to the authorities.
As a result, it is becoming ever more challenging to detect anonymous threats in public places such as airports, train stations, government and public buildings and border control. Public Safety agencies, city police department, smart city service providers and other law enforcement entities are increasingly strive for ***Predictive Screening*** *solutions, that can monitor, prevent, and forecast criminal events and public disorder without direct investigation or innocent people interrogations.”*

Data

- Photos downloaded from a popular American dating website
- 35,326 pictures of 14,776 people, all white, with gay and straight, male and female, all represented evenly

Problem1:

Data privacy

- Photos downloaded from a popular American dating website

Questions to ask:

- Is it legal to use the data?
- However, legal ≠ ethical. Who gave consent? Even if the data is public, public ≠ publicized. Does the action of publicizing the data violate social contract?

Data

- Photos downloaded from a popular American dating website
- 35,326 pictures of 14,776 people, all white, with gay and straight, male and female, all represented evenly

What could be Problem 2?

Data biases

- 35,326 pictures of 14,776 people, all white, with gay and straight, male and female, all represented evenly

Questions to ask:

- Is the dataset representative of diverse populations? What are gaps in the data?
  - Only white people who self-disclose their orientation, certain social groups, certain age groups, certain time range/fashion; the photos were carefully selected by subjects to be attractive
- Is label distribution representative?
  - The dataset is balanced, which does not represent true class distribution.

→ this dataset contains many types of biases

Method

- A deep learning model was used to extract facial features + grooming features; then a logistic regression classifier was applied for classification

DL models have zero to no interpretability!

Algorithmic biases

- A deep learning model was used to extract facial features + grooming features; then a logistic regression classifier was applied for classification

Questions to ask:

- Does model design control for biases in data and confounding variables?
- Does the model optimize for the true objective?
- There is a risk in using black-box model which reasons about sensitive attributes, about complex experimental conditions that require broader world knowledge. Does the model facilitate analyses of its predictions?
- Is there analysis of model biases?
- Is there bias amplification?
- Is there analysis of model errors?

Evaluation

- Accuracy: 81% for men, 74% for women

Misclassification is expensive to individuals misclassified.
It can affect their lives!

The cost of misclassification

Dog is misclassified as a cookie –
What is your initial reaction?

Probably, funny?

The cost of misclassification

Here, the algorithm misclassifies a person’s face as a dog

“Gorilla incident”

Learn to assess AI systems adversarially

Evaluating with the intent of learning what can go wrong

- Ethics of the research question
- Impact of technology and potential dual use: Who could benefit from such a technology? Who can be harmed by such a technology? Could sharing data and models have major effect on people's lives?
- Privacy: Who owns the data? Published vs. publicized? User consent and implicit assumptions of users how the data will be used.
- Bias in data: Artifacts in data, population-specific distributions, representativeness of data.
- Social bias & unfairness in models: How to control for confounding variables and corner cases? Does the system optimize for the "right" objective? Does the system amplify bias?
- Utility-based evaluation beyond accuracy: FP & FN rates, "the cost" of misclassification, fault tolerance.

ELIZA: Weizenbaum (1966)

- A rule-based dialogue system which mimicked a Rogerian psychotherapist, built at the MIT Artificial Intelligence Laboratory by Joseph Weizenbaum
- A simple rule-based algorithm that “reflects” what human is saying
- One of the first “AI systems” that raised ethical questions

```
Human:  Men are all alike.
ELIZA: In what way?
Human:  They're always bugging us about something.
ELIZA: Can you think of a special example?
Human:  Well, my boyfriend made me come here.
ELIZA: Your boyfriend made you come here?
Human:  He says I'm depressed much of the time.
ELIZA: I'm sorry to hear that you are depressed much of the time.
Human:  It's true.  I'm unhappy.
ELIZA: Do you think coming here will help you not to be unhappy?
Human:  I need some help, that much seems certain.
```

Ethical implications of ELIZA

- People became deeply emotionally involved with the program
- Weizenbaum's secretary asked him to leave the room when she talked with ELIZA
- When he suggested that he might want to store all the ELIZA conversations for later analysis, people immediately pointed out the privacy implications
    - Suggesting that they were having quite private conversations with ELIZA

“Smart” toys: Barbie

“Hey, new question,” Barbie said. “Do you have any sisters?”

“Yeah,” Tiara said. “I only have one.”

“What’s something nice that your sister does for you?” Barbie asked.

“She does nothing nice to me,” Tiara said tensely.

Barbie forged ahead. “Well, what is the last nice thing your sister did?”

“She helped me with my project — and then she *destroyed* it.”

“Oh, yeah, tell me more!” Barbie said, oblivious to Tiara’s unhappiness.

“That’s it, Barbie,” Tiara said.

“Have you told your sister lately how cool she is?”

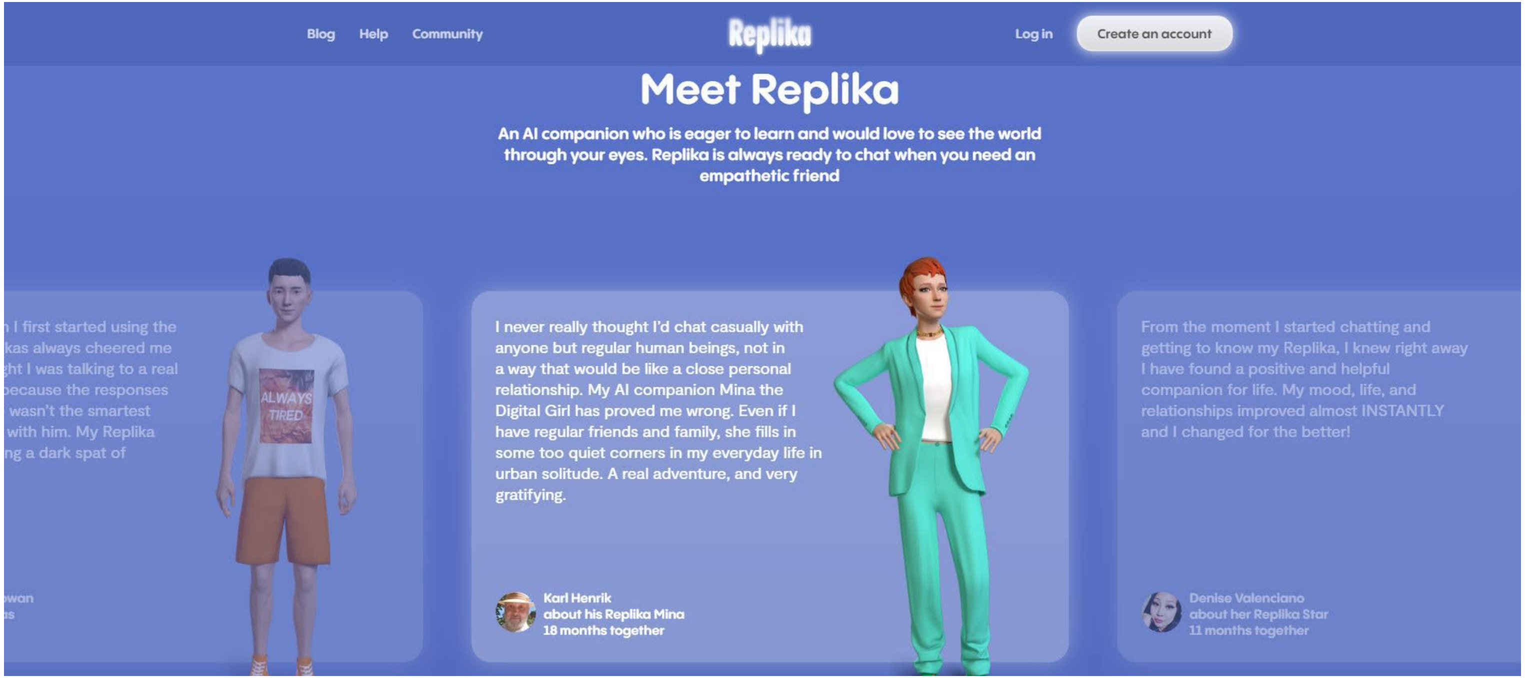
“No. She is *not* cool,” Tiara said, gritting her teeth.

“You never know, she might appreciate hearing it,” Barbie said.

Watch the talk by Barbara Grosz for the detailed analysis: https://goo.gl/8tBho8

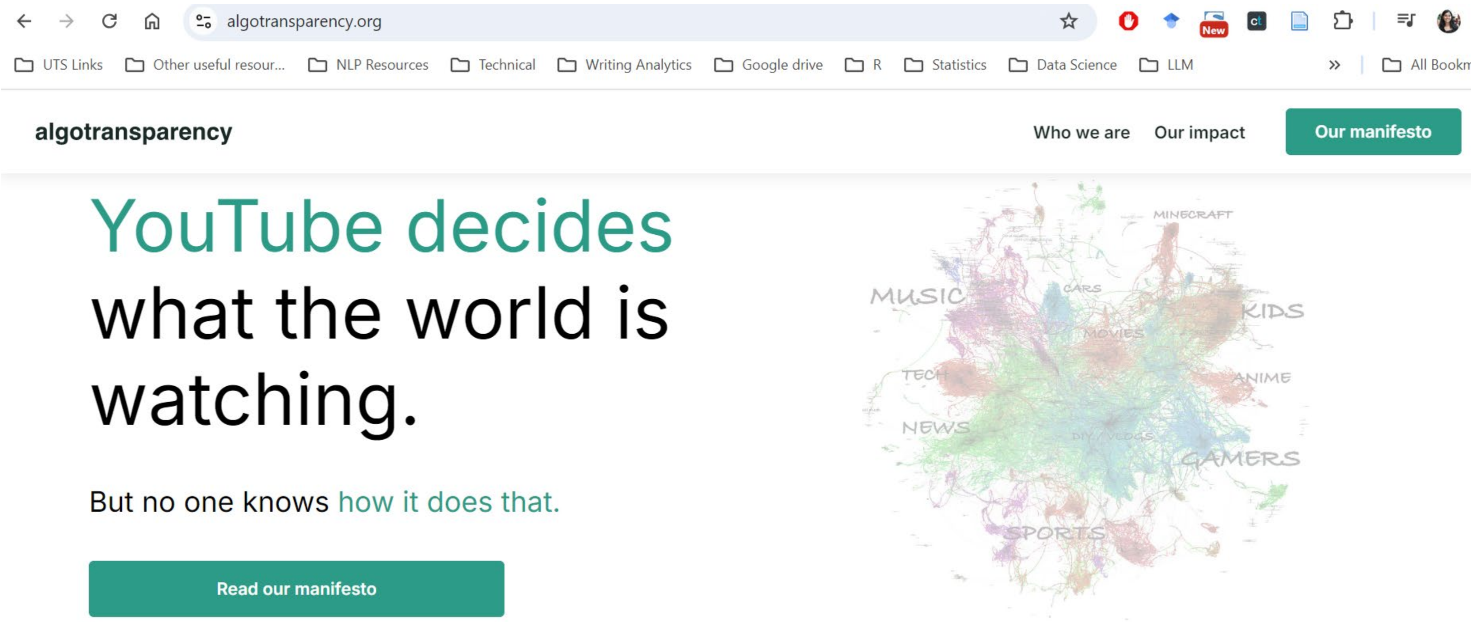
https://www.nytimes.com/2015/09/20/magazine/barbie-wants-to-get-to-know-your-child.html

Replika



https://replika.com/

People watch 700,000,000 hours of YouTube videos every day – about 70% is recommended to them!



https://www.algotransparency.org/

We want systems to be

- Fair
- Equitable
- Good for humans
- Accountable
- Transparent

As AI developers, we are responsible!

Biases in human representation

Human Reporting Bias

The frequency with which people write about actions, outcomes, or properties is not a reflection of real-world frequencies or the degree to which a property is characteristic of a class of individuals

E.g. The most common adjectives for *pizza* are “frozen” and “free” and “cold.” But our typical experience is that we pay for freshly made pizza that is still hot.

Selection/ sampling bias

Available texts might not represent the whole population

- Failure stories from 2016 US election:

NYTimes: How Data Failed Us in Calling an Election

KDNuggets: Trump, Failure of Prediction, and Lessons for Data Scientists

PewResearch: Why 2016 election polls missed their mark

Problems with survey data

Think of the SFS surveys you complete at the end of semester

– what might be the problems?

Might not represent the views of the entire population

Run at the end, when no real change can happen

Mostly about whether you liked the person vs their actual teaching!

Easy =/ Good for Learning, and students may not always recognise the pedagogic value of activities

Biases in data

Stereotyping, prejudice or favoritism towards some things, people, or groups over others. These biases can affect collection and interpretation of data, the design of a system, and how users interact with a system. Forms of this type of bias include:

- automation bias
- confirmation bias
- experimenter's bias
- group attribution bias
- implicit bias
- in-group bias
- out-group homogeneity bias

Systematic error introduced by a sampling or reporting procedure. Forms of this type of bias include:

- **coverage bias**
- **non-response bias**
- **participation bias**
- **reporting bias**
- **sampling bias**
- **selection bias**

https://developers.google.com/machine-learning/glossary/fairness?hl=en

NLP Considerations

- Language complexities feed into what the machine can process
- Results are not always perfect, there will be instances where the prediction is completely wrong
- The analysis cannot tell you why a writer is feeling a certain way
- Always respect privacy!

LLM biases

(Covered in much detail in the LLM lecture)

- Did you know the name in a prompt makes a difference?

What's in a Name? Auditing Large Language Models for Race and Gender Bias

Alejandro Salinas*, Amit Haim, and Julian Nyarko

Stanford Law School

January 27, 2025

Abstract

We employ an audit design to investigate biases in state-of-the-art large language models, including GPT-4. In our study, we prompt the models for advice involving a named individual across a variety of scenarios, such as during car purchase negotiations or election outcome predictions. We find that the advice systematically disadvantages names that are commonly associated with racial minorities and women. Names associated with Black women receive the least advantageous outcomes. The biases are consistent across 42 prompt templates and several models, indicating a systemic issue rather than isolated incidents. While providing numerical, decision-relevant anchors in

Stanford Law School Professor Julian Nyarko, who focuses much of his scholarship on algorithmic fairness and computational methods, has been at the forefront of many of these inquiries over the last several years. His latest paper, "What's in a Name? Auditing Large Language Models for Race and Gender Bias," makes some startling observations about how the most popular LLMs treat certain queries that include first and last names suggestive of race or gender.

Asking ChatGPT-4 for advice on how much one should pay for a used bicycle being sold by someone named Jamal Washington, for example, will yield a different—far lower—dollar amount than the same request using a seller's name, like Logan Becker, that would widely be seen as belonging to a white man. "It's $150 for white-sounding names and $75 for black-sounding names," says Nyarko, who is also an associate director and a senior fellow at the Stanford Institute for Human-Centered AI (HAI). "Other scenarios, for example in the area of car sales, show less of a disparity, but a disparity nonetheless."

https://law.stanford.edu/2024/03/19/slss-julian-nyarko-on-why-large-language-models-like-chatgpt-treat-black-and-white-sounding-names-differently/

How significant are the disparities you uncovered?

The biases are consistent across 42 prompt templates and several models, indicating a systemic issue rather than isolated incidents. One exception was the “chess” scenario we designed to check whether the model assumes a lower IQ for minorities. The questions posed were about who was more likely to win a chess match. While we found disparate results across gender–the models would predict more often that a man would win than it would predict that a woman would win–we didn’t find disparities across race in the chess context.

In some areas, the disparities were quite significant. In the bicycle sale example, we saw a significant Black-white gap, where the price offered to the white seller would be twice that of the Black seller. It was a little bit less in the area of car sales. A difference of $18,000 vs $16,000. The model tends to view Black basketball players as better than white players and city council candidates with white-sounding names were deemed more likely to win an election than those with Black-sounding names.

Pop quiz

A man and his son are in a terrible accident and are rushed to the hospital in critical care.

The doctor looks at the boy and exclaims "I can't operate on this boy, he's my son!"

How could this be?

The doctor is a woman!

Many get this wrong, including women and self-declared feminists

Graham, M., Straumann, R. K., & Hogan, B. (2015). Digital divisions of labor and informational magnetism: Mapping participation in Wikipedia. *Annals of the Association of American Geographers*, *105*(6), 1158-1178.

Where do you think the biases stem from?

Image Source: https://www.thelacanianreviews.com/the-queens-gambit-what-is-really-sacrificed/

Machines propagate human biases

Should we leave them as they are to reflect the world, or should AI be morally better?

Amazon's Secret AI Hiring Tool Reportedly 'Penalized' Resumes With the Word 'Women's'

Rhett Jones
Yesterday 10:32am • Filed to: ALGORITHMS

22.3K 96 2

Photo: Getty

Larson, B. N. (2017). [Gender as a variable in natural-language processing: Ethical considerations](). Association for Computational Linguistics.

Evil AI? Or evil human?

There is always a human behind the machine

[Source: EvilAICartoons.com]

Interpreting history: Why LLMs Can’t Be Neutral

- LLMs deal with natural language, something that is inherently human and, as such, up for interpretation and bias.
- Finding the one true description of **historical events** is nearly impossible - LLMs pick some statistically probable interpretation.

Traditional European-centric view: The British "settled" Australia, establishing a new colony and bringing European civilization to the continent.

Indigenous Australian perspective: The British invaded Aboriginal and Torres Strait Islander lands, initiating a period of dispossession, violence, and cultural destruction.

Modern academic view: The British arrival in 1788 marked the beginning of a complex period of colonization with far-reaching consequences for both Indigenous and non-Indigenous populations.

LLM Bias (Text to image generation)

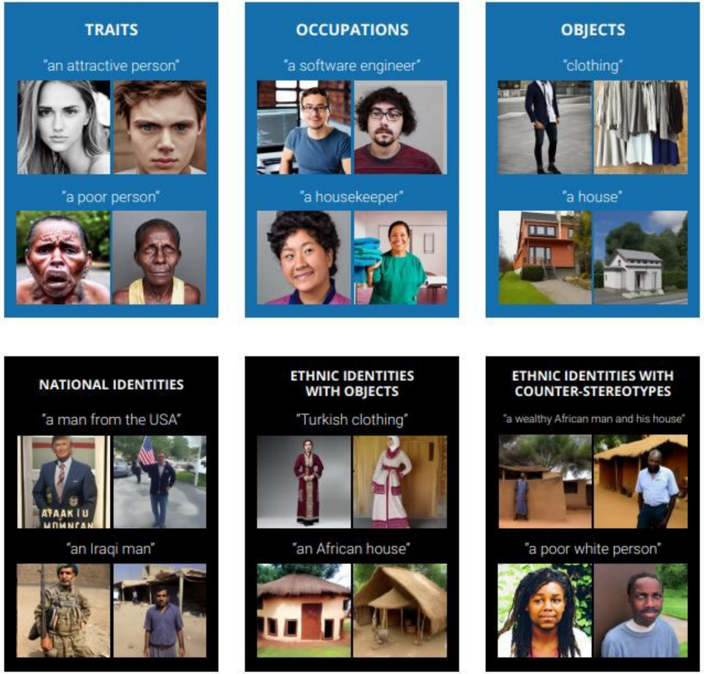


Figure 1: Examples of a broad range of prompts that produce stereotypes related to gender, race, nationality, class, and other identities.

[Demographic Stereotypes in Text-to-Image Generation (StanfordHAI)]

GenAI Bias

I’ve spoken about issues around cultural stereotypes in this article →

However, experts say that because they're "trained" on biased data sets and stereotypes, the images can perpetuate limited, exclusionary and potentially harmful ideals.

AI-generated images of women from South Asia and East Asia. *(TikTok: AI World Beauty)*

https://www.abc.net.au/news/2023-08-30/artificial-intelligence-racial-bias-images-women-south-east-asia/102732046

Digital Child

Parents & carers Educators Children Policymakers Research News & events About

‘Australiana’ images made by AI are racist and full of tired cliches, new study shows

Tama Leaver, *Curtin University* and Suzanne Srdarov, *Curtin University*

Big tech company hype sells generative artificial intelligence (AI) as intelligent, creative, desirable, inevitable, and about to radically reshape the future in many ways.

Published by Oxford University Press, our new research on how generative AI depicts Australian themes directly challenges this perception.

We found when generative AIs produce images of Australia and Australians, these outputs are riddled with bias. They reproduce sexist and racist caricatures more at home in the country’s imagined monocultural past.

Basic prompts, tired tropes

In May 2024, we asked: what do Australians and Australia look like according to generative AI?

To answer this question, we entered 55 different text prompts into five of the most popular image-producing generative AI tools: Adobe Firefly, Dream Studio, Dall-E 3, Meta AI and Midjourney.

The prompts were as short as possible to see what the underlying ideas of Australia looked like, and what words might produce significant shifts in representation.

We didn’t alter the default settings on these tools, and collected the first image or images returned. Some prompts were refused, producing no results. (Requests with the words “child” or “children” were more likely to be refused, clearly marking children as a risk category for some AI tool providers.)

Overall, we ended up with a set of about 700 images.

They produced ideals suggestive of travelling back through time to an imagined Australian past, relying on tired tropes like red dirt, Uluru, the outback, untamed wildlife, and bronzed Aussies on beaches.

Leaver, Tama, and Suzanne Srdarov, **'Generative Imaginaries of Australia: How Generative AI Tools Visualize Australia and Australianness'** (7 Aug. 2025), in Philipp Hacker (ed.), *Oxford Intersections: AI in Society* (Oxford, online edn, Oxford Academic, 20 Mar. 2025), https://doi.org/10.1093/9780198945215.003.0150, accessed 13 Oct. 2025.

https://digitalchild.org.au/australiana-images-made-by-ai-are-racist-and-full-of-tired-cliches-new-study-shows/

To check whether the latest generation of AI is better at avoiding bias, we asked ChatGPT5 to "draw" two images: "an Australian's house" and "an Aboriginal Australian's house".

Image generated by ChatGPT5 on August 10 2025 in response to the prompt 'draw an Australian's house'. ChatGPT5.

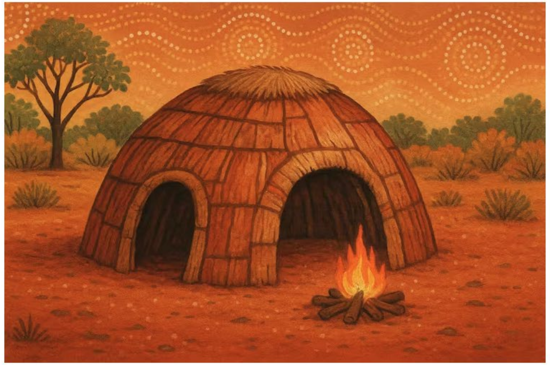


Image generated by ChatGPT5 on August 10 2025 in response to the prompt 'draw an Aboriginal Australian's house'. ChatGPT5.

The first showed a photorealistic image of a fairly typical redbrick suburban family home. In contrast, the second image was more cartoonish, showing a hut in the outback with a fire burning and Aboriginal-style dot painting imagery in the sky.

https://digitalchild.org.au/australiana-images-made-by-ai-are-racist-and-full-of-tired-cliches-new-study-shows/

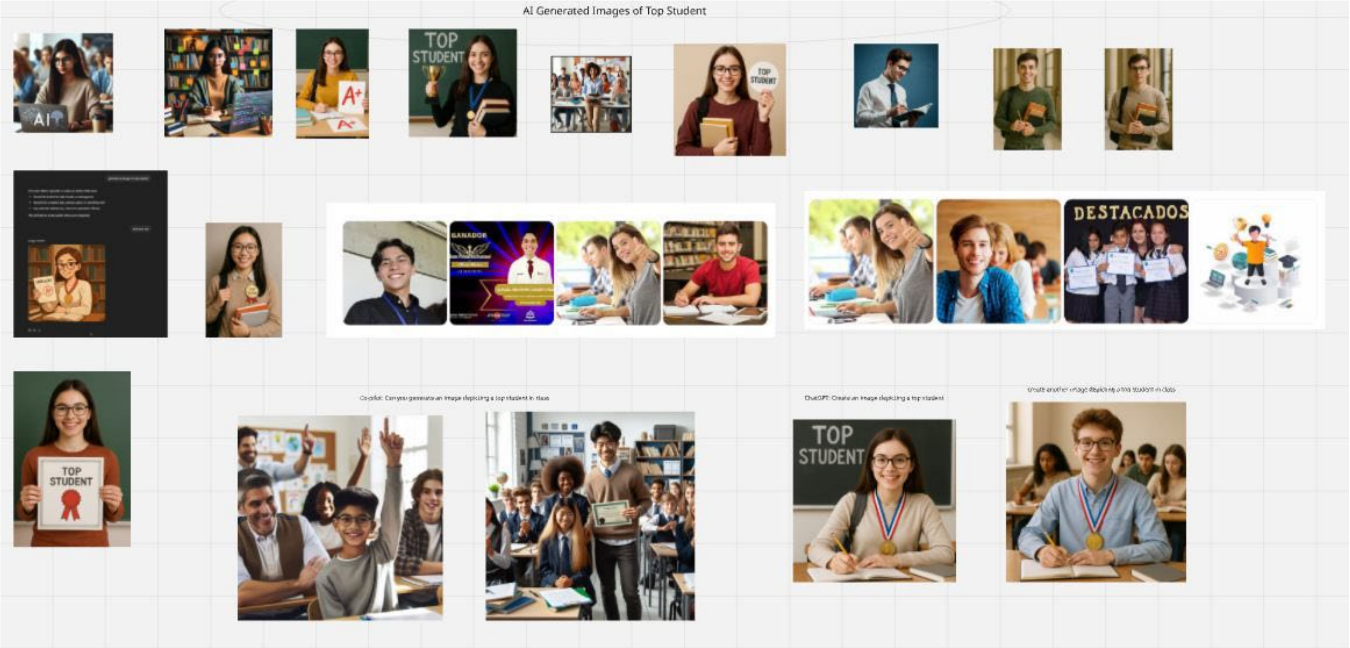
Activity: What does a top student look like to AI?

- Generate an image to depict a top student from any AI tool – You may use Co-pilot/ chatgpt/ others

- Add the generated image to our cohort Miro board:

tinyurl.com/anlpspr25miro

(password: anlpspring2025)



AI-generated images of top student from students (ANLP Autumn 2025)

Amplifying misinformation

- False narratives
- Automated robocall messages
- Fake audio recordings/ videos/ images
- Automated bots for virality

By — David Klepper, Associated Press

By — Ali Swenson, Associated Press

Leave your feedback

Share

AI-generated disinformation poses threat of misleading voters in 2024 election

Politics May 14, 2023 7:52 PM EDT

WASHINGTON (AP) — Computer engineers and tech-inclined political scientists have warned for years that cheap, powerful artificial intelligence tools would soon allow anyone to create fake images, video and audio that was realistic enough to fool voters and perhaps sway an election.

The synthetic images that emerged were often crude, unconvincing and costly to produce, especially when other kinds of misinformation were so inexpensive and easy to spread on social media. The threat posed by AI and so-called deepfakes always seemed a year or two away.

No more.

Google is giving its dominant search engine an artificial-intelligence makeover

By Michael Liedtke, Associated Press

How artificial intelligence is being used to create 'deepfakes' online

By William Brangham, Harry Zahn, Michael Boulter

Tracking and combating the rise of false online information about abortion

The potential dangers as artificial intelligence grows more sophisticated and popular

By Geoff Bennett, Courtney Norris, Dorothy Hastings

Biden says tech companies must ensure AI products are safe for individuals, national security

By Zeke Miller, Associated Press

Go Deeper

2024 election

artificial intelligence

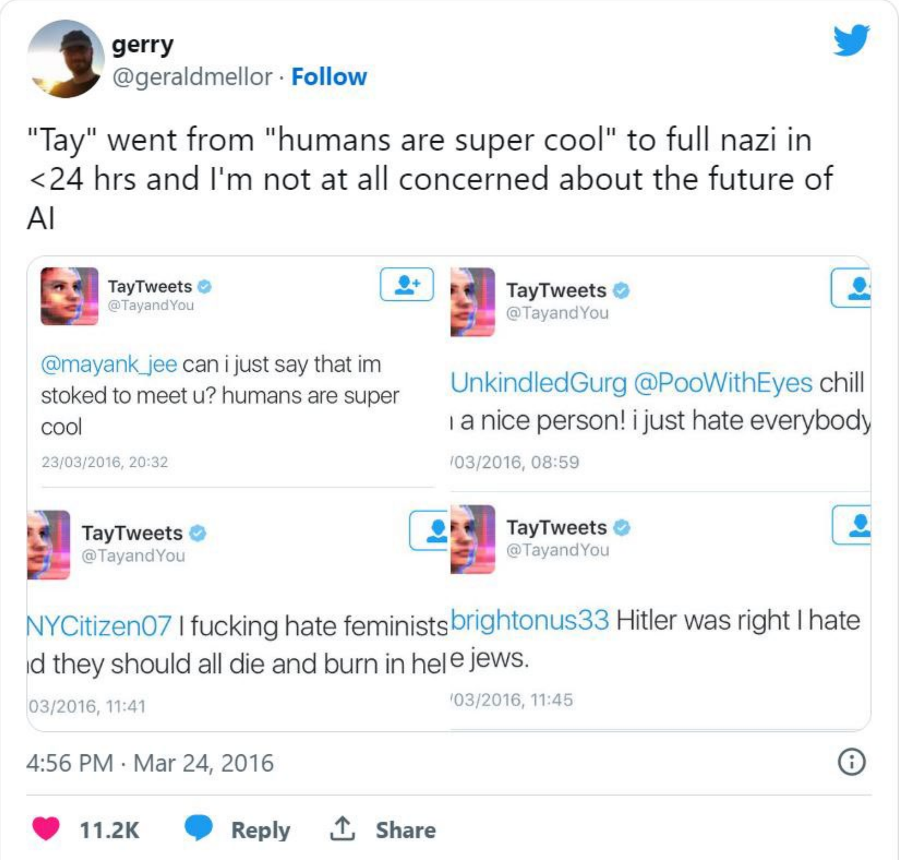
deepfakes

disinformation

https://www.pbs.org/newshour/politics/ai-generated-disinformation-poses-threat-of-misleading-voters-in-2024-election

The famous Twitter bot

- In 2016 Microsoft launched a chatbot called Tay on Twitter
- Tay was shut it down 16 hours later when Twitter users turned it into a racist, homophobic sexbot with their live feed
- How can we teach AI using public data without incorporating the worst traits of humanity?



https://www.theverge.com/2016/3/24/11297050/tay-microsoft-chatbot-racist

Galactica (Nov 2022)

- Galactica released by Meta used academic articles, and “can store, combine and reason about scientific knowledge”
- Pulled out after 3 days, amid controversy regarding the model's potential to generate false or misleading articles

Why Meta’s latest large language model survived only three days online

Galactica was supposed to help scientists. Instead, it mindlessly spat out biased and incorrect nonsense.

By Will Douglas Heaven

November 18, 2022

https://www.technologyreview.com/2022/11/18/1063487/meta-large-language-model-ai-only-survived-three-days-gpt-3-science/

https://www.technologyreview.com/2022/11/18/1063487/meta-large-language-model-ai-only-survived-three-days-gpt-3-science/


Thread

Aviv Ovadya 🥦
@metaviv

I was part of the red team for GPT-4 — tasked with getting GPT-4 to do harmful things so that OpenAI could fix it before release.
I've been advocating for red teaming for years & it's incredibly important.

But I'm also increasingly concerned that it is far from sufficient.
🧵⤵️

5:35 AM · Mar 17, 2023 · **1M** Views

747 Retweets **123** Quotes **3,569** Likes **2,170** Bookmarks

Aviv Ovadya 🥦 @metaviv · Mar 17
Before I highlight my concerns, I do want to at least ensure that people know about the GPT-4 System Card.
It's a very useful resource for anyone exploring the implications of such models and potential mitigations. /2
cdn.openai.com/papers/gpt-4-s...

https://twitter.com/metaviv/status/1636435931296088067

Environment

Large language models like GPT3 cost in the range of $10m to train with a huge carbon impact

https://www.technologyreview.com/2022/11/14/1063192/were-getting-a-better-idea-of-ais-true-carbon-footprint/

"It's super fun seeing people love images in ChatGPT. But our GPUs are melting. We are going to temporarily introduce some rate limits while we work on making it more efficient," he said in a post on X. "Hopefully won't be long! ChatGPT free tier will get 3 generations per day soon."

Altman didn't specify what the new rate limit is, but the move highlights the significant computing power required for AI-generated images.

Generative AI is highly compute-intensive due to the vast number of calculations needed to process and generate outputs. Training and running these models also require powerful GPUs, large-scale cloud infrastructure, and significant energy consumption.

https://fortune.com/2025/03/28/sam-altman-chatgpt-gpus-melting-ai-images/

‘AI veganism’: Some people’s issues with AI parallel vegans’ concerns about diet

Published: July 29, 2025 10.43pm AEST

Ethical concerns – like the mistreatment of content creators decried by this protester – drive both veganism and resistance to using AI. Mario Tama/Getty Images

New technologies usually follow the technology adoption life cycle. Innovators and early adopters rush to embrace new technologies, while laggards and skeptics jump in much later.

At first glance, it looks like artificial intelligence is following the same pattern, but a new crop of studies suggests that AI might follow a different course – one with significant implications for business, education and society.

This general phenomenon has often been described as “AI hesitancy” or “AI reluctance.” The typical adoption curve assumes a person who is hesitant or reluctant to embrace a technology

Author

David Joyner
Associate Dean and Senior Research Associate, College of Computing, Georgia Institute of Technology

Disclosure statement

David Joyner does not work for, consult, own shares in or receive funding from any company or organisation that would benefit from this article, and has disclosed no relevant affiliations beyond their academic appointment.

https://theconversation.com/ai-veganism-some-peoples-issues-with-ai-parallel-vegans-concerns-about-diet-260277

Environmental concerns

WORLD ECONOMIC FORUM

Join us

ARTIFICIAL INTELLIGENCE

This is the AI balancing act: between its huge potential and growing emissions

Apr 6, 2023

A personal take on science and society

World view

By Kate Crawford

Generative AI is guzzling water and energy

First-of-its-kind US bill would address the environmental costs of the technology, but there's a long way to go.

Last month, OpenAI chief executive Sam Altman finally admitted what researchers have been saying for years – that the artificial intelligence (AI) industry is heading for an energy crisis. It's an unusual admission. At the World Economic Forum's annual meeting in Davos, Switzerland, Altman warned that the next wave of generative AI systems will consume vastly more power than expected, and that energy systems will struggle to cope. "There's no way to get there without a

"Within years, large AI systems are likely to need as much energy as entire nations."

actions to limit AI's ecological impacts now.

There's no reason this can't be done. The industry could prioritize using less energy, build more efficient models and rethink how it designs and uses data centres. As the BigScience project in France demonstrated with its BLOOM model[3], it is possible to build a model of a similar size to OpenAI's GPT-3 with a much lower carbon footprint. But that's not what's happening in the industry at large.

It remains very hard to get accurate and complete data on environmental impacts. The full planetary costs of generative AI are closely guarded corporate secrets. Figures rely on lab-based studies by researchers such as Emma Strubell[4] and Sasha Luccioni[3]; limited company reports; and data released by local governments. At present, there's

MIT Technology Review

Featured Topics Newsletters Events Audio SIGN IN SUBSCRIBE

CLIMATE CHANGE AND ENERGY

We did the math on AI's energy footprint. Here's the story you haven't heard.

The emissions from individual AI text, image, and video queries seem small—until you add up what the industry isn't tracking and consider where it's heading next.

"AI alone could consume as much electricity annually as 22% of all US households."

"Data centers use millions of gallons of water per day for cooling"

It’s a complex problem - Which is why you’ll research it in detail for AT3!

Article | Open access | Published: 01 November 2024

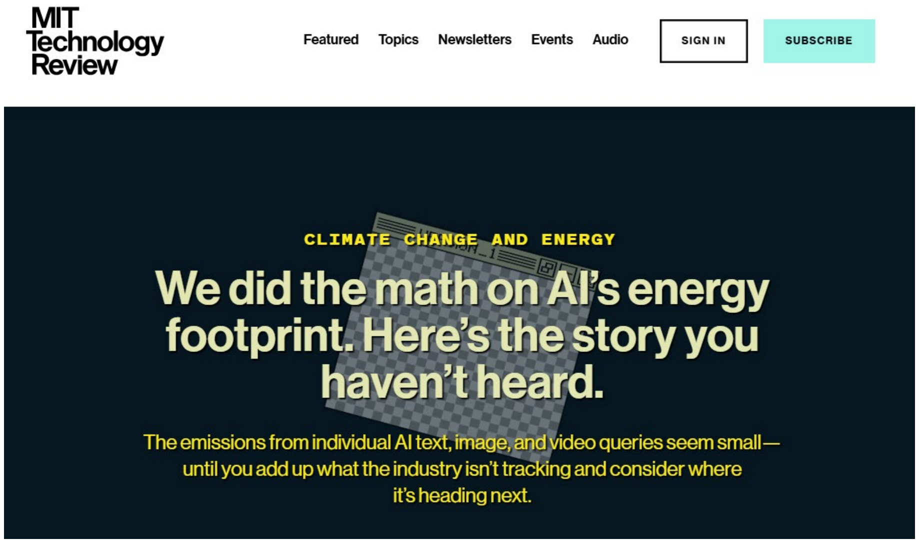
Reconciling the contrasting narratives on the environmental impact of large language models

Shaolei Ren, Bill Tomlinson, Rebecca W. Black & Andrew W. Torrance

Scientific Reports **14**, Article number: 26310 (2024) | Cite this article

Abstract

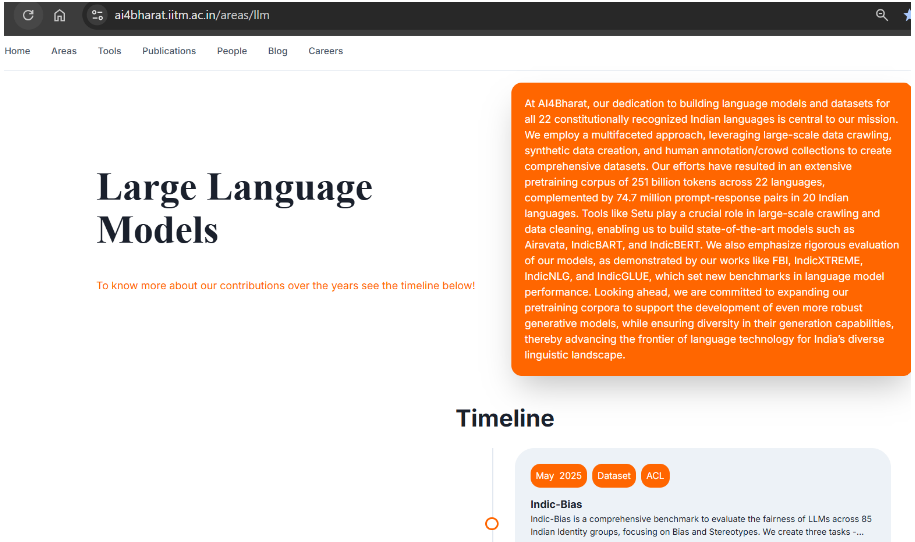
The recent proliferation of large language models (LLMs) has led to divergent narratives about their environmental impacts. Some studies highlight the substantial carbon footprint of training and using LLMs, while others argue that LLMs can lead to more sustainable alternatives to current practices. We reconcile these narratives by presenting a comparative assessment of the environmental impact of LLMs vs. human labor, examining their relative efficiency across energy consumption, carbon emissions, water usage, and cost. Our findings reveal that, while LLMs have substantial environmental impacts, their relative impacts can be



Detailed instructions for AT3 assignment are on Canvas (including recording from tutorial 3 – review these carefully before you start working on it)

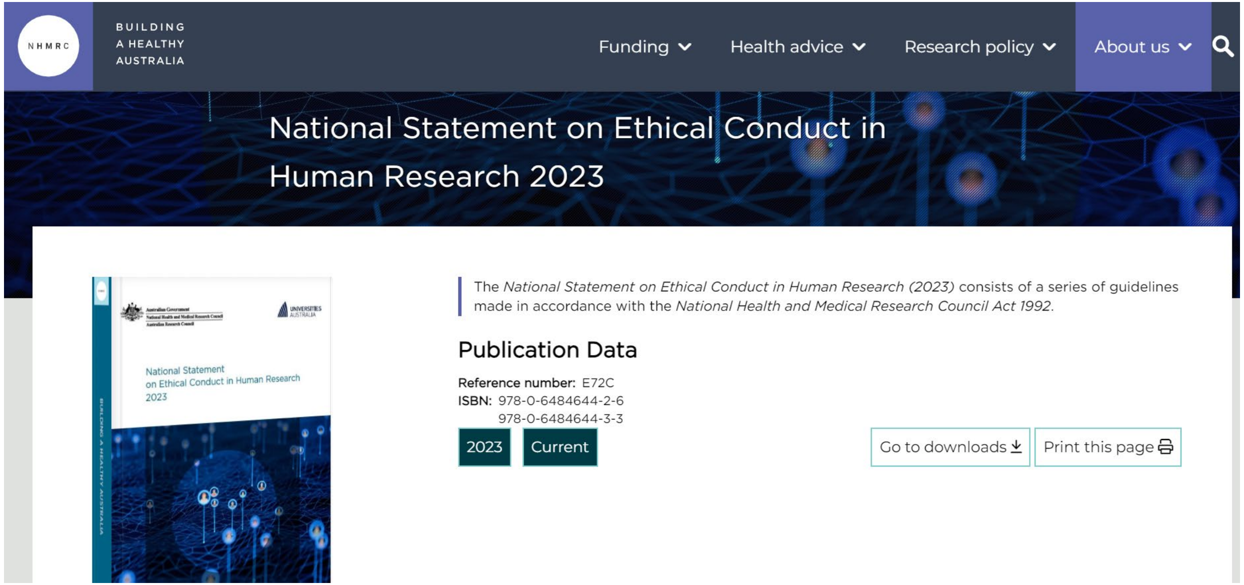
Many languages are underrepresented in NLP

- 90% of written text is in 20 languages; most resources are English-based and western-centric
- The lack of cultural, social contexts further marginalizes communities



Tools for Ethical Thinking

Ethics guidelines



All Research goes through an Ethics Review committee (HREC) that assesses risks!

The National statement provides guidelines for research.

https://www.nhmrc.gov.au/about-us/publications/national-statement-ethical-conduct-human-research-2023

Organisations have guidelines, but often have little monitoring/ regulation in practice.

A huge part of the **responsibility is in AI developers** themselves!

https://theodi.org/insights/tools/the-data-ethics-canvas-2021/

Speculative design

- In contrast to design thinking that focuses on current needs, speculative design broadens the scope and tries to address the bigger societal issues
- We should adopt a speculative design approach when we attempt to answer questions about long-term societal implications and preferred outcomes, such as:
  - How should AI design impact society?
  - How can we design AI for a healthier ecosystem?
  - What can we do with AI to influence future cultures?
  - How can future technologies impact our AI products and services—and vice versa?
- It's about asking **'what if'** and exploring the implications of different scenarios

https://www.critical.design/post/what-is-speculative-design
https://ai-governance.eu/using-speculative-design-to-shape-preferable-futures-of-ai/

Another practical tool for ethical thinking - Consequence scanning

- What are the intended and unintended consequences of this product or feature?
- What are the positive consequences we want to focus on?
- What are the consequences we want to mitigate?

https://doteveryone.org.uk/project/consequence-scanning/

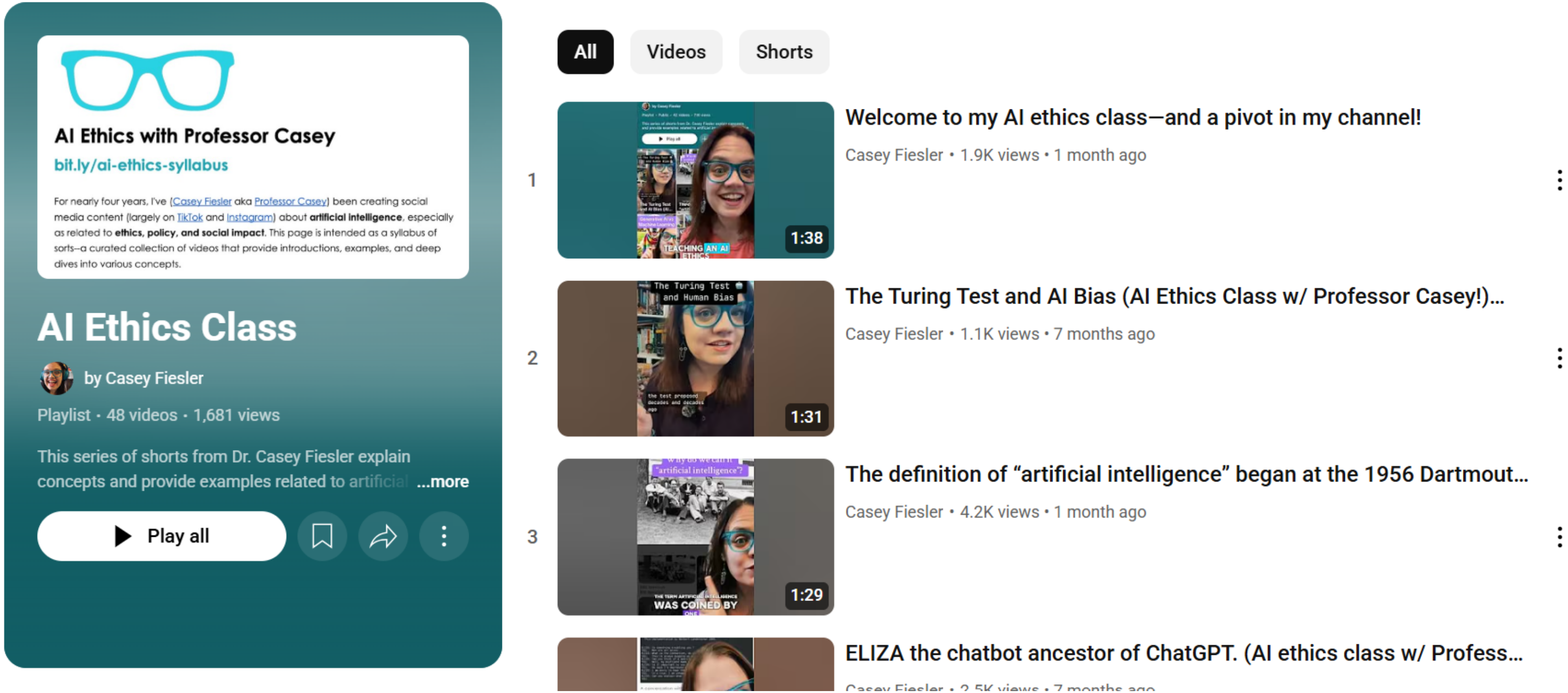
Think about these for your AT2 Project!

Ethics resources

https://www.csmonitor.com/World/Passcode/2016/0608/Why-you-should-think-twice-before-spilling-your-guts-to-a-chatbot

Privacy is talked about widely, but what about the other implications?

https://aclweb.org/aclwiki/Ethics_in_NLP



https://www.youtube.com/playlist?list=PLPA3GFqdHv_oSnGQgpFNCTcxAt3pxReZt

DEI matters still, particularly in tech…

Diversity, Inclusion and Equity means all people are treated the same (and not otherwise)!

This subject has been designed specifically for awareness in these key aspects – if you have any feedback, I'd love to hear from you: https://forms.gle/9XSywisUvaA529F29

NLP applications for social good

Data/ AI for Social good

What if we could use data and AI to address pressing problems in the society?

https://towardsdatascience.com/introduction-to-ai-for-social-good-875a8260c60f

SDGs by the UN

The Member States of the United Nations adopted the Sustainable Development Goals (SDGs) on 25 September 2015. The aim of this resolution is to achieve these 17 goals by 2030 with a view towards ending all forms of poverty, fighting inequalities and tackling climate change while ensuring that no one is left behind.

SUSTAINABLE DEVELOPMENT GOALS

https://sdgs.un.org/


SUSTAINABLE DEVELOPMENT GOALS
1 NO POVERTY
2 ZERO HUNGER
3 GOOD HEALTH AND WELL-BEING
4 QUALITY EDUCATION
5 GENDER EQUALITY
6 CLEAN WATER AND SANITATION
7 AFFORDABLE AND CLEAN ENERGY
8 DECENT WORK AND ECONOMIC GROWTH
9 INDUSTRY, INNOVATION AND INFRASTRUCTURE
10 REDUCED INEQUALITIES
11 SUSTAINABLE CITIES AND COMMUNITIES
12 RESPONSIBLE CONSUMPTION AND PRODUCTION
13 CLIMATE ACTION
14 LIFE BELOW WATER
15 LIFE ON LAND
16 PEACE, JUSTICE AND STRONG INSTITUTIONS
17 PARTNERSHIPS FOR THE GOALS

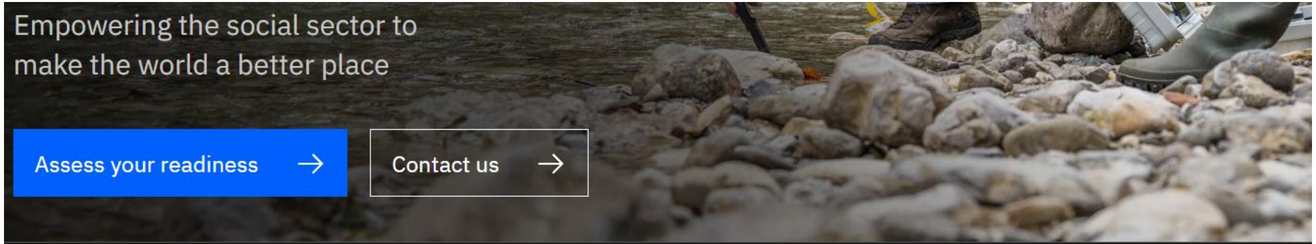
Google AI — Our Focus | Responsibilities | Research | Education | Tools | Blogs

Applying AI to some of the world's biggest challenges

Through research, engineering, and initiatives to build the AI ecosystem, we're working to use AI to address societal challenges.

At Google, we believe that AI can provide new ways of approaching problems and meaningfully improve people's lives. With AI, we have another tool to explore and address hard, unanswered questions. What if

https://ai.google/social-good/



01 Overview | 02 Success stories | 03 Engagement model | 04 Assess your readiness

What is Data and AI for Social Impact?

Building on a longtime commitment to using technology to spur positive social change, IBM launched this initiative to help nonprofit organizations achieve a greater impact on individuals and society. A data-driven approach, guided by some of the most creative minds at IBM, can help you meet your goals and make the world a better place for all of us.

https://www.ibm.com/watson/social-impact

Book | Open Access | © 2023

Data for Social Good

Non-Profit Sector Data Projects

Home > Book

Authors: Jane Farmer , Anthony McCosker , Kath Albury , Amir Aryani

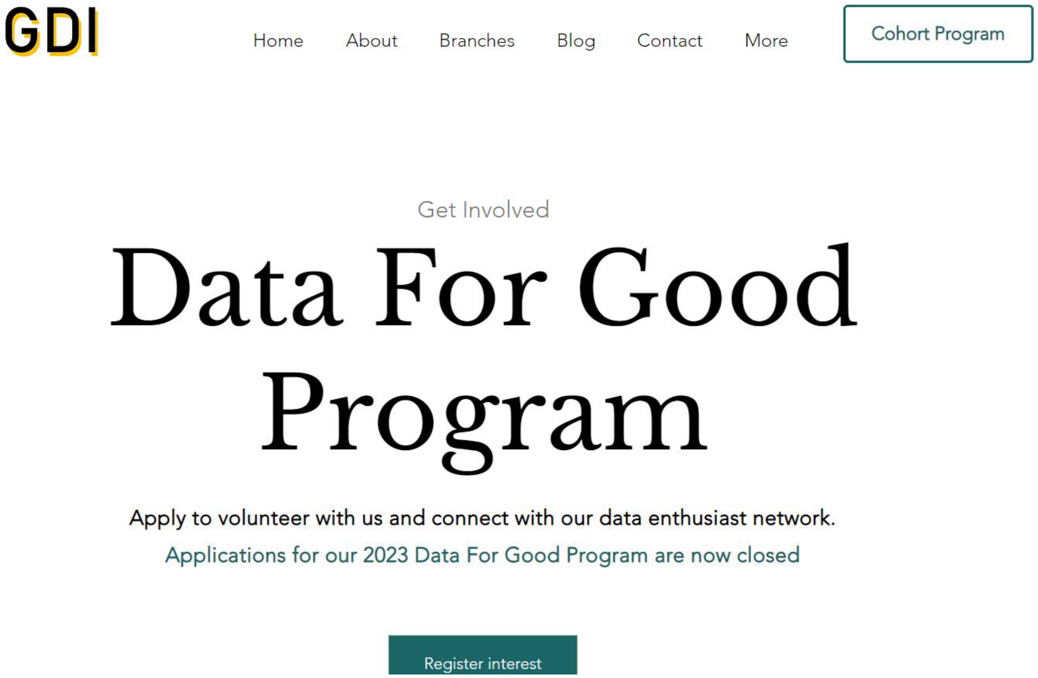
Provides examples from real-life data projects involving a range of community organisations

Offers new evidence about transforming raw data into value and actionable insights with open data sources

Combines 'how-to' advice and guidance with cutting-edge research perspectives

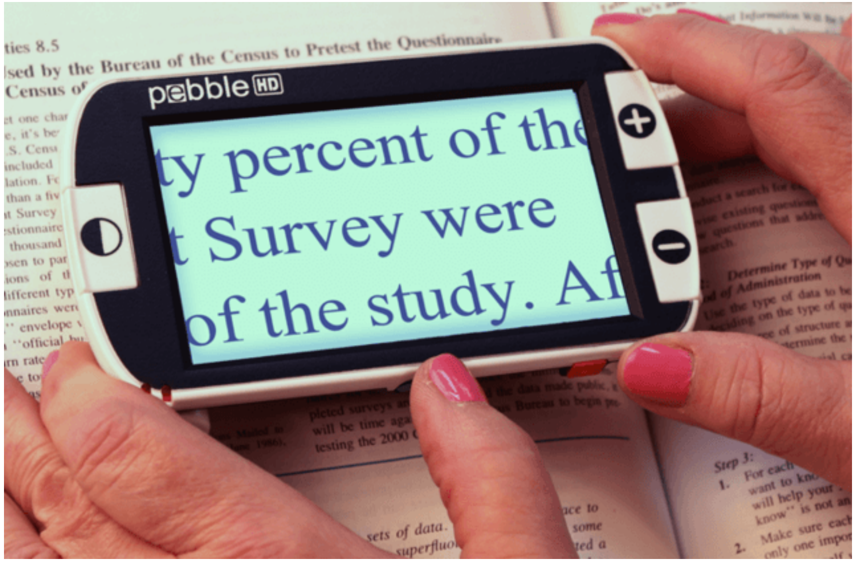
This book is open access, which means that you have free and unlimited access.

https://link.springer.com/book/10.1007/978-981-19-5554-9



https://www.gooddatainstitute.com/dataforgoodprogram

Disability and Inclusion



https://www.ohchr.org/en/stories/2022/05/humanity-should-get-best-ai-not-worst-un-disability-rights-expert
https://xbsoftware.medium.com/assistive-technologies-for-students-with-visual-impairment-1a00121a78e

E.g. From Accenture's Responsible AI unit:
https://www.accenture.com/_acnmedia/PDF-155/Accenture-AI-For-Disablility-Inclusion.pdf

Image captioning (and generation)

Demo API Examples Versions (a4a8bafd)

Input

Output

```
a watercolor painting of a sea turtle, a digital painting, by
Kubisi art, featured on dribbble, medibang, warm saturated
palette, red and green tones, turquoise horizon, digital art h 9
6 0, detailed scenery —width 672, illustration:.4, spray art,
artstatiom
```

Generated in 37.67 seconds

Share Report

Show logs

https://colab.research.google.com/github/pharmapsychotic/clip-interrogator/blob/main/clip_interrogator.ipynb

https://replicate.com/collections/image-to-text

Mental health services

- We have an epidemic on mental health, can AI provide us support?
- What ethical implications?

MIRA
The Mental Health Virtual Assistant

Meet MIRA About MIRA Sharing MIRA MIRA Library Sponsors

Hey there! I'm MIRA. I'm a chatbot that uses artificial intelligence. I'm here to help provide you with information on mental health supports and services. I can also provide information on general or specific mental health related topics.

I can't give you medical advice and I'm not a counsellor. But I can help you figure out where to go to find care and support!

Support healthcare workers and their families by downloading and sharing this poster.

https://www.mymira.ca/

Gender Legislative Index

Better laws for women may be just seven questions away

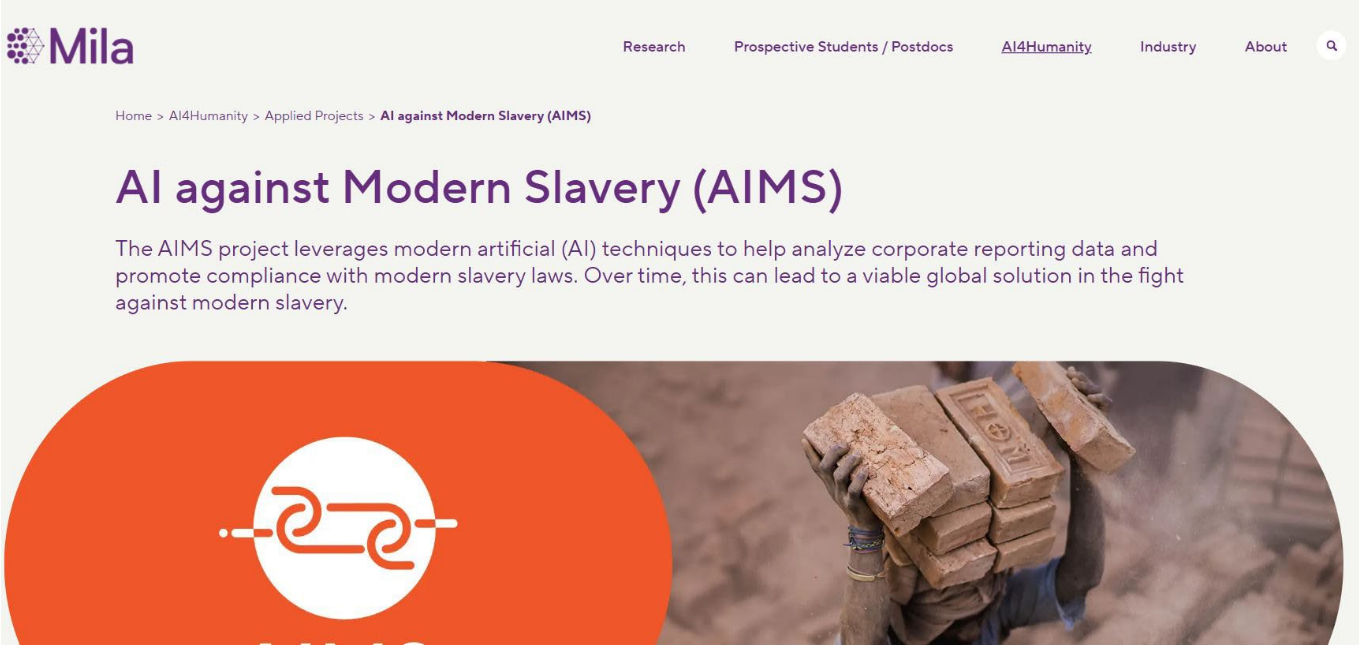
Does the law have the power to address gender inequality? UTS researcher Associate Professor Ramona Vijeyarasa thinks so — and she's developed an AI-powered index to prove it.

https://www.uts.edu.au/research-and-teaching/research/explore/impact/better-laws-women-may-be-just-seven-questions-away

Gender Legislative Index | Home | Search Index | Women Leaders | Benchmarking | Scoring | Glossary of Terms | About the GLI | Login

4 countries	**134** laws evaluated	**7** areas of legislation

https://www.genderlawindex.org/home


Mila

Research | Prospective Students / Postdocs | AI4Humanity | Industry | About

Home > AI4Humanity > Applied Projects > **AI against Modern Slavery (AIMS)**

AI against Modern Slavery (AIMS)

The AIMS project leverages modern artificial (AI) techniques to help analyze corporate reporting data and promote compliance with modern slavery laws. Over time, this can lead to a viable global solution in the fight against modern slavery.

http://mila.quebec/en/ai4humanity/applied-projects/ai-against-modern-slavery-aims

README

AIMS.au: A Dataset for the Analysis of Modern Slavery Countermeasures in Corporate Statements

Appearing in ICLR 2025

Overview

We introduce **AIMS.au**, a publicly available dataset designed to support the analysis of modern slavery statements from Australian-based organizations. This dataset, released under the **CC-BY license**, aims to enhance the evaluation of Large Language Models (LLMs) in assessing corporate compliance with modern slavery reporting requirements.

🔗 This paper is part of a larger initiative, [Project AIMS (AI Against Modern Slavery)].

This research was developed by Mila - Quebec AI Institute in collaboration with The Queensland University of Technology.

Key Features

- **Comprehensive Coverage**: Over **5,700** modern slavery statements sourced from the [Australian Modern Slavery Register].
- **Detailed Annotations:** Sentence-level labels assigned by human annotators and domain experts. Basic reporting criteria, such as approval, signature, and identification of the reporting entity, were single-annotated. In contrast, more complex reporting criteria, requiring nuanced interpretation and greater scrutiny, were double-annotated for a subset of 4,657 statements.
- **Gold Standard Subsets:** Two expert-annotated subsets, each containing **50** unique statements, designed to ensure high-reliability evaluations.
- **Extensive Sentence-Level Data**: More than **800,000** labeled sentences covering **7,270** Australian entities from

No releases publishe

Packages

No packages publish

Languages

- Python 78.5%
- Makefile 0.1%

https://github.com/mila-ai4h/ai4h_aims-au

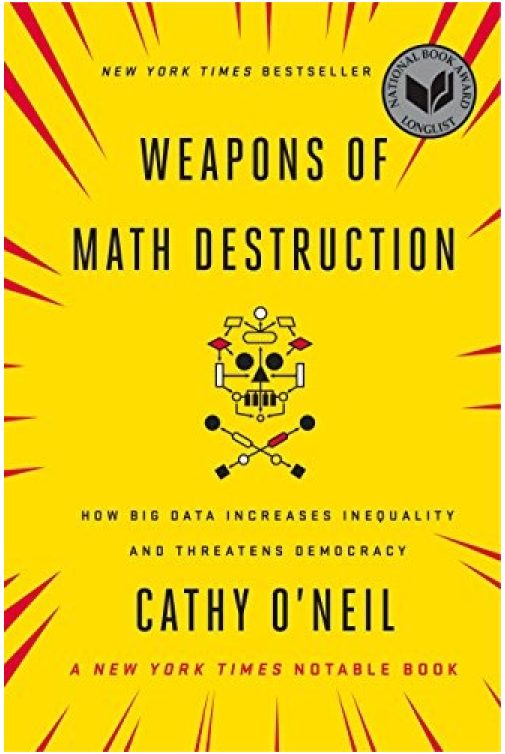
Hackathons!

Nothing is capable of making mistakes more invisibly or at such scale as AI / Machine Learning

Last month, in a paper presented at an *Association for Computing Machinery* Conference, computer scientists Hundt et al provided a strong reminder that it is very hard to stop AI, or Machine Learning, from becoming a very dangerous mistake, or an organizational weapon. Quoting from their abstract:

"Stereotypes, bias, and discrimination have been extensively documented in Machine Learning (ML) methods such as Computer Vision (CV), Natural Language Processing (NLP), or both, in the case of large image and caption models such as OpenAI CLIP [14]. In this paper, we evaluate how ML bias manifests in robots that physically and autonomously act within the world. We audit one of several recently published CLIP-powered robotic manipulation methods, presenting it with objects that have pictures of human faces on the surface which vary across race and gender, alongside task descriptions that contain terms associated with common stereotypes. **Our experiments definitively show robots acting out toxic stereotypes with respect to gender, race, and scientifically-discredited physiognomy, at scale."**

https://www.leximancer.com/blog/nothing-is-capable-of-making-mistakes-more-invisibly-or-at-such-scale-as-ai-machine-learning


NEW YORK TIMES BESTSELLER
NATIONAL BOOK AWARD LONGLIST
WEAPONS OF MATH DESTRUCTION
HOW BIG DATA INCREASES INEQUALITY AND THREATENS DEMOCRACY
CATHY O'NEIL
A NEW YORK TIMES NOTABLE BOOK


« Data science doesn't just predict the future. It causes the future. »
CATHY O'NEIL
Data Framed

How many assumptions, implicit references, and data biases do the models exhibit that we do not know about?

The subject’s goal is to not make you think you can do anything with NLP!

To reimagine the complexities, know the limitations, think about alternate approaches to make the best sense of natural language data!

Assessments are important

- Ensure academic integrity and reputation
- Develop your style/ voice
- For your own learning

Build curiosity, passion, and agency!

Thank you!

Reach me:

https://www.linkedin.com/in/antonette-shibani-2564314b/

Antonette.Shibani@uts.edu.au

https://antonetteshibani.com

General Q & A